

PULSEFIT

Multi-Facility Fitness Membership & Attendance Tracking Platform

1. Problem Statement

Fitness centers face challenges in managing member information, diverse membership plans, renewals, and attendance across multiple facilities. Manual systems can lead to delays, errors, and poor user experience at entry points.

2. Project Overview

PulseFit is a smart microservices-based platform that digitizes membership management, subscription lifecycle, and attendance tracking with a secure, scalable, and high-availability architecture.

3. Key Objectives

- Manage member profiles and membership plans
- Handle subscriptions and renewals
- Track gym attendance and facility usage
- Support multiple facilities
- Provide secure JWT-based authentication
- Enable service discovery and scalable inter-service communication.

4. Design Thinking & User Research

We conducted a survey with 50+ gym-goers to understand their needs and challenges. Key issues included long waiting times at gym entry, lack of clarity about membership status, difficulty tracking attendance, renewal-management challenges, and the need for access across multiple facilities.

5. System Architecture

The platform follows a microservices architecture consisting of Auth Service, Member Service, Subscription Service, Attendance Service, API Gateway, and Eureka Server. The API Gateway provides routing, security, and load balancing, while Eureka supports service registration and discovery.

6. Microservices Overview

Auth Service handles authentication and JWT token management. Member Service manages registration and profile details. Subscription Service manages plans, subscriptions, and renewals. Attendance Service tracks check-ins, check-outs, and facility usage. API Gateway routes requests and applies security and load balancing.

7. Data Handling & Integration

Each service maintains responsibility for its domain data, supporting loose coupling and independent service management. Member, subscription, attendance, and authentication data are handled by their respective services. REST APIs/OpenFeign can be used for service-to-service communication.

8. Innovative Ideas

Potential enhancements include smart facility-usage analytics, automatic renewal reminders, multi-facility access, real-time attendance tracking, and insights for better facility management.

9. Technology Stack

Spring Boot • Spring Cloud • Microservices Architecture • JWT Authentication • Netflix Eureka • API Gateway • REST APIs • OpenFeign • Database Integration

10. Team

Padala Gowtham Avinash Reddy – Team Lead (2400033275)

Palla Gnana Sai Kiran – Team Member (2400031822)

Gade Revanth – Team Member (2400033232)

11. Conclusion

PulseFit demonstrates how microservices can be applied to a real-world fitness management scenario. The solution focuses on modularity, secure access, service discovery, scalable communication, and an improved membership and attendance experience across multiple facilities.

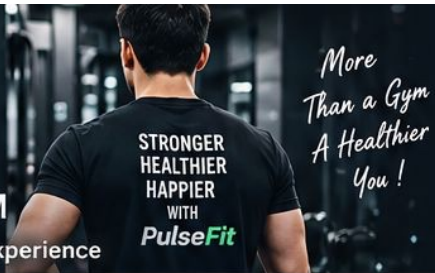
PulseFit Project Poster



FITNESS TODAY A HEALTHIER TOMORROW

A SMART MICROSERVICES-BASED FITNESS MEMBERSHIP & ATTENDANCE MANAGEMENT PLATFORM

Manage Memberships | Track Attendance | Multiple Facilities | Better Fitness Experience



1. Team Details



Padala Gowtham Avinash Reddy
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2. Problem Statement

Fitness centers face challenges in managing member information, diverse membership plans, renewals, and attendance across multiple facilities. Manual systems can lead to delays, errors, and poor user experience at entry points.

"How might we make gym access simple, smart and seamless for every member across all facilities?"

3. Our Solution

PulseFit is a microservices-based platform that digitizes membership management, subscription lifecycle, and attendance tracking with secure, scalable, and high-availability architecture.



Manage Members



Membership Plans & Renewals



Track Attendance



Multi-Facility Insights

4. Survey & Key Insights

We conducted a survey with **50+** gym-goers to understand their needs and challenges.

Common Issues Identified

- Long waiting time at gym entry
- Lack of clarity on membership status
- No easy way to track attendance
- Difficulty in managing renewals
- Need for access across multiple facilities

5. Empathy Map (Based on Survey)

Says

- "I don't want to wait at the entrance."
- "I want to know my membership status."

Thinks

- Wants quick access
- Needs transparency in plans and renewals
- Wants a better gym experience

Does

- Visits gym regularly
- Checks membership details
- Uses multiple facilities

Feels

- Frustrated by long queues
- Worried about expiry
- Motivated to stay fit
- Wants convenience

6. Persona (From a Real Survey)



Name: Rahul
Age: 22
Occupation: Student
Goal: Attend gym regularly without delays
Problems:

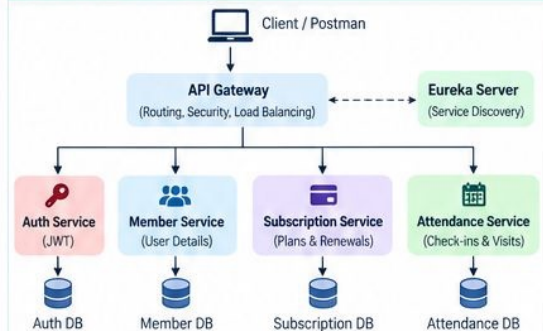
- Not aware of expiry dates
- Manual entry causes delays
- Difficult to track attendance
- Uses multiple facilities

Need: A simple and smart system to manage membership and gym access.

7. Customer Journey Map



8. System Architecture



9. Microservices Overview

	Auth Service	Handles user authentication and JWT token management.
	Member Service	Manages member registration and profile details.
	Subscription Service	Handles membership plans, subscriptions and renewals.
	Attendance Service	Tracks gym check-ins, check-outs and facility usage.
	API Gateway	Routes requests, applies security and load balancing.
	Eureka Server	Service discovery and registration.

10. Data Handling & Integration

Member Data	Subscription Data	Attendance Data
Member ID	Subscription ID	Attendance ID
Name	Member ID	Member ID
Email	Plan	Facility
Phone	Start Date	Check-in
Facility	Expiry Date	Check-out
	Status	Date

Service Communication Flow

Attendance Service → Subscription Service → Member Service
(Using REST APIs / OpenFeign)

11. Innovative Ideas

- Smart facility usage analytics (peak hours)
- Automatic renewal reminders
- Multi-facility access for members
- Real-time attendance tracking
- Insights for better facility management

"Empowering healthier lives through smarter technology."

*Fit People
Build Better Tomorrows*



More Fitness



More Health



Happier Communities



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